

CAIR2 Integration - Quick Reference

Aeliusmd | ClaudMD | HL7 VXU v2.5.1

Goal: Send vaccination records from ClaudMD clinics to CAIR2 (California Immunization Registry) using HL7 v2.5.1 VXU messages over SOAP.

1. Architecture (30-second view)

```
Nurse saves vaccine (EHRVaccines, IsVaccine=1)
|
Queue row (outbox or IsSubmitted=0)
|
clair_service.py (every 5 min, worker pool)
|
Build HL7 VXU -> SOAP submitSingleMessage -> CAIR2
|
ACK: AA=success | AE=retry | AR=failed
```

2. Your Environment

Item	Value
Org code (Aeliusmd)	SF-013259
Master DB	ClaudMD_QA_Setup@10.103.0.201
Activation key (example)	20000002 -> ClaudMD_VCOMC_QA_2
Clinic registry table	dbo.ClinicSetup
CAIR contact	CAIRDataExchange@cdph.ca.gov

3. CAIR SOAP Endpoints

Env	URL
ONBOARDING (test now)	cdph-interop-stage.cdph.ca.gov/.../client_ServiceHttpSoap12Endpoint
WSDL (stage)	cdph-interop-stage.cdph.ca.gov/CASFC-WS/IFService?WSDL
PRODUCTION	cdph-interop-prod.cdph.ca.gov/.../client_ServiceHttpSoap12Endpoint
SOAP operation	submitSingleMessage (VXU)

4. Sample HL7 VXU Message (from CAIR email)

```
MSH|^~\&||CA0012345||CAIR2|20260816023022-0700||VXU^V04^VXU_V04|
{GUID}|P|2.5.1|||AL|AL|||Z22^CDCPHINVS|CA0054321
PID|1||AC10293^^CA0012345^MR||Alvarez^Maria|...|19880412|F|...
|^NET^Internet^maria@example.com~^PRN^CP^^^916^5550134|...
ORC|RE|48213^CMC|48213^CMC|...
RXA|0|1|20260810||00069-1000-03^^NDC|0.3|mL^^UCUM||
00^NEW IMMUNIZATION RECORD^NIP001|^CA0054321||FA7205||PFR^^MVX|||CP|A
OBX|1|CE|64994-7^...^LN|1|V01^Not VFC eligible^HL70064||||F|||20260810
```

5. HL7 Segments - What Each Does

Segment	Purpose	Example
MSH	Message header	VXU^V04^VXU_V04, facility ID, AL/AL
PID	Patient demographics	Name, DOB, sex, address, PID-13 phone/email
PD1	Privacy/consent	Protection indicator Y/N
ORC	Order control	RE, check-in ID
RXA	Vaccine given	NDC/CVX, dose, lot, date, CPTA
RXR	Route & site	IM, left arm (optional)
OBX	VFC eligibility	V01=not VFC eligible

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6. ClaudMD Database Mapping (Real Tables)

CAIR / Excel field	ClaudMD table.column
Patient ID (PID-3)	Patients.AccountNumber
Check-in ID (ORC-2/3)	CheckinsHeader.Id
Vaccination record	EHRVaccines
Vaccine code (RXA-5)	ServiceCodes.NDCNumber (CVX missing)
Dose (RXA-6)	EHRVaccines.Dosage
Lot (RXA-15)	EHRVaccines.LotNumber
Manufacturer (RXA-17)	EHRVaccines.Manufacturer (need MXX code)
Admin date (RXA-3)	EHRVaccines.VaccineDate + VaccineTime
Phone/email (PID-13)	Patients.CellPhone, HomePhone, Email
Already submitted?	EHRVaccines.IsSubmitted
Filter real vaccines	EHRVaccines.IsVaccine=1

7. PID-13 Contact - REQUIRED by CAIR

CAIR rejects/warns if phone AND email are both missing. At least one must be sent in PID-13.

Home phone: ^PRN^PH^^^213^5555555
 Cell phone: ^PRN^CP^^^916^5550134
 Email: ^NET^Internet^patient@email.com

All together (separated by ~):

^PRN^PH^^^213^5555555~^NET^Internet^email@x.com~^PRN^CP^^^916^5551234

QA issue: most Patients have CellPhone but Email is NULL.

8. Vaccine Codes

Code	Used for	Example
CVX	Vaccine type (clinical)	08=Hep B, 207=COVID
MXX	Manufacturer	PFR=Pfizer, MSD=Merck
NDC	Product package	00069-1000-03
CPT	Billing only	NOT used in CAIR VXU primary field

BLOCKER: ServiceCodes has no CvxCode or MvxCode columns yet. Must add per CAIR email requirements before reporting works.

9. Outbox / Worker Flow

Status flow:

PENDING -> PROCESSING -> SENT
 -> RETRY (network/timeout, retry later)
 -> FAILED (bad data, manual fix)

Worker query (one queue for new + retries):

WHERE status='PENDING'
 OR (status='RETRY' AND next_retry_at <= now)

Multi-clinic: 3 workers, 1 batch per clinic at a time,
 shared pool (not 1 worker per clinic forever).

10. MSH Key Fields Cheat Sheet

Field	Value	Notes
MSH-4	SI-013259	Your CAIR org code
MSH-6	CAIR2	Receiving facility
MSH-9	VXU V04 VXU V04	Message type
MSH-10	GUID	Unique per message

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MSH-11

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P or T

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P=production, T=training

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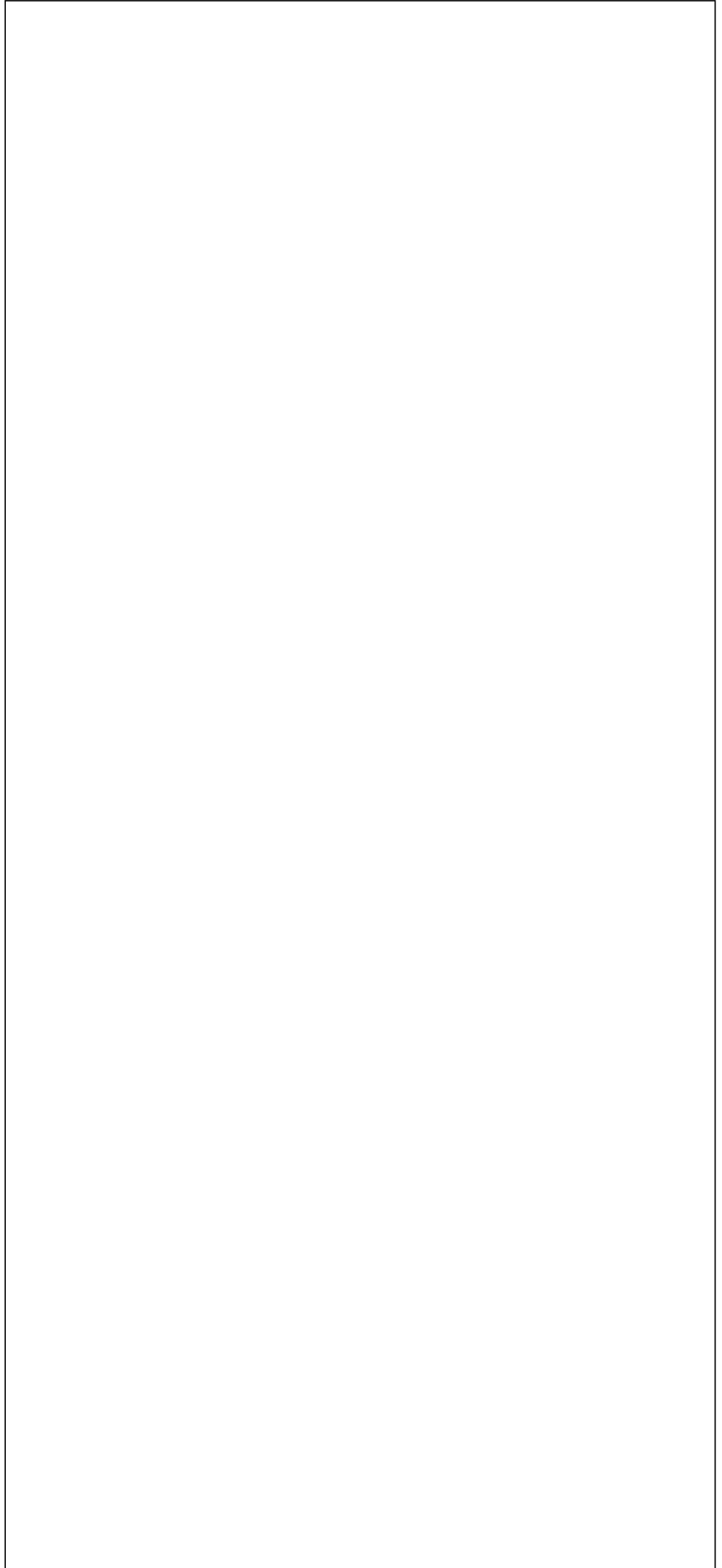
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MSH-15/16 MSH-15/16	AL / AL	Always get ACK
MSH-22 MSH-22	Site ID	Where vaccine was given

11. RXA Key Fields Cheat Sheet

Field	Value	Notes
RXA-1/2 RXA-1/2	0 / 1	Fixed counters
RXA-3 RXA-3	XXXXMMDD	Vaccine date
RXA-5 RXA-5	NDC or CVX	One only, not both
RXA-6 RXA-6	0-3 or 999	999 if dose unknown
RXA-7 RXA-7	mL or UGUM	Unit if dose known
RXA-9.1 RXA-9.1	00	00=given shot (not historical)
RXA-15 RXA-15	FA7205	Lot number
RXA-20/21 RXA-20/21	CP / A	Complete / Add
OBX-5 OBX-5	V01	VFC eligibility code

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12. Project Files

File	Purpose
cair_service.py	Background worker - run continuously
demo_build_vxu.py	Test HL7 generation without DB
cair_integration/hl7/vxu_builder.py	Builds HL7 message
cair_integration/cair/soap_client.py	Sends to CAIR SOAP
cair_integration/worker/coordinator.py	Multi-clinic dispatcher
.env	DB + CAIR credentials (do not commit)
CAIR2_HL7v2.5.1DataExchangeSpecs.pdf	Official CAIR spec
cair_emails.txt	Endpoints, sample HL7, PID-13 rules

13. Blockers Before Go-Live

1. Add CvxCode + MvxCode to ServiceCodes (CDC lookup tables)
2. Collect patient email or cell phone (PID-13 required)
3. Fill lot number + route on vaccine entry
4. Create test data: EHRVaccines with IsVaccine=1 (QA has 0 today)
5. Test on STAGE endpoint, email CAIR team for review
6. Switch to PROD endpoint after approval

14. Quick Test Commands

```
pip install -r requirements.txt
python demo_build_vxu.py          # print sample HL7
python cair_service.py            # start background worker
```